



Research paper

Kinome analyses of *Candida albicans*, *C. parapsilosis* and *C. tropicalis* enable novel kinases as therapeutic drug targets in candidiasis

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ABSTRACT

Candida spp. have attracted considerable attention as they cause serious human diseases in immunocompromised individuals. The genomes of the pathogenic *Candida* spp. have been sequenced, but systemic characterizations of their kinomes are yet to be reported. As in various eukaryotes, the protein kinases play crucial regulatory roles in pathogenicity of *Candida*. Increased frequency of antifungal resistance in *Candida* spp. requires significant attention to explore novel therapeutic molecules for their control. The present *in-silico* study involves novel bioinformatics strategies to identify the kinase proteins and their potential drug targets with the purpose to combat fungal infections. The study reports 103, 107 and 106 kinase proteins from 3 *Candida* spp., *C. albicans*, *C. parapsilosis* and *C. tropicalis*, respectively. Moreover, 79 common kinase proteins were identified, of which 54 proteins play essential roles in *Candida* spp. and 42 proteins were human non-homologues. Among the essential and human non-homologous protein kinases, 9 were found to be common essential human non-homologues, of which 6 are uniquely present in *Candida*. These 6 protein kinases namely, Hsl1, Npr1, Ptk2, Kin2, Ksp1 and orf19.3854 (CAALFM_CR06040WA) are involved in various molecular and cellular processes regulating virulence or pathogenicity. Further, these 6 kinases are prioritized as potential drug targets and explored for discovering new lead compounds against candidiasis. The drug repurposing approach for these 6 kinases show 13 approved drugs and investigational compounds that might play substantial inhibitory roles during combating candidiasis.

1. Introduction

Candidiasis is an opportunistic, superficial and systemic infection of humans that is caused by *Candida* spp. (Antinori et al., 2016; López-Martínez, 2010). *Candida* spp. are regarded as the fourth leading cause of bloodstream infections in hospitals, resulting in the increased frequency of candidiasis every year (Wisplinghoff et al., 2004). The prognosis of candidiasis is generally poor, which leads to mortality rate of 30%–60% (Ben-Ami, 2018). In candidiasis, *C. albicans* is predominantly diagnosed followed by *C. tropicalis*, which is the second most virulent species (Zuza-Alves et al., 2017) and *C. parapsilosis* is a significant neonatal pathogen, associated with 10% mortality (Pammi et al., 2013).

In India, these 3 species are primarily involved in candidiasis. *C. tropicalis* is the most frequently found invasive *Candida* species in India and Pakistan (Tan et al., 2015; Xess et al., 2007). In India, extensive investigation of candidiasis showed 6.5 person per 1000 patient in hospital getting infected with a mortality rate of 45% (Chakrabarti et al., 2015).

Protein kinases (PKs) are the largest group among the family of enzymes found in organisms. The kinases catalyze the most important protein phosphorylation processes. In *Candida*, the phosphorylation of proteins, switch many cellular processes including signal transduction, metabolism, growth and development, cell cycle progression and responses to environmental stimuli (Alonso Monge et al., 2006; Biswas

Abbreviations: *C. albicans*, *Candida albicans*; *C. parapsilosis*, *Candida parapsilosis*; *C. tropicalis*, *Candida tropicalis*; *S. cerevisiae*, *Saccharomyces cerevisiae*; PKs, Protein kinases; ePKs, Eukaryotic protein kinase; aPKs, atypical protein kinase; N-terminal, Amino terminal; C-terminal, Carboxy terminal; AGC, Protein kinase A, G, and C; CAMK, Calcium and calmodulin-regulated kinase group; CMGC, Glycogen synthase kinases and CDK-like kinases; CK1, Casein kinase 1; STE, Sterile kinase; RGC, Receptor guanylate cyclases; TK, Tyrosine kinase; TKL, Tyrosine kinase-like; 3D structure, Three dimensional structures.

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et al., 2007; Correia et al., 2017, 2010; Cottier and Mühlischlegel, 2009; Das et al., 2019; Kumamoto, 2005; Ramírez-Zavala et al., 2017). For the versatile essential roles of these PKs, they are considered to be perspective disease control targets. PKs can be divided into eukaryotic protein kinase (ePKs) and atypical protein kinase (aPKs). The ePKs are primarily composed of tyrosine kinase and serine/ threonine kinase. The ePKs have highly conserved motif containing 11 subdomains with N-terminal lobe comprising of antiparallel β -sheets and a C-terminal lobe with α -helices (Manning et al., 2002). Hanks and Hunter classified protein kinases into 8 subfamilies, including AGC, CAMK, CMGC, CK1, STE, RGC, TK, TKL and “Other group” (Miranda-Saavedra and Barton, 2007). Protein kinases regulate several aspects of cellular life, making it one of the most important group for drug targets (Cohen, 2002). Recent studies have revealed that the antifungal resistance against *C. albicans* is enhanced worldwide. (Castanheira et al., 2016; Pfaller et al., 2010). In India, about 12% and 2% of isolated pathogens showed resistance to azole and multi drugs, respectively (Chakrabarti et al., 2015). Therefore, it is becomes essential to identify the novel kinase proteins, which would be necessary for pathogen survival and human non-homologues, that could be the potential therapeutic targets. We applied a robust bioinformatics pipeline to identify and classify the kinase proteins. The classified protein kinases were further prioritized through drug repurposing workflow to identify the essential kinase proteins and their therapeutic drug targets (Fig. 1).

2. Material and methods

2.1. Definition and prediction of *C. albicans*, *C. parapsilosis* and *C. tropicalis* kinomes

We executed proteome wide kinome analysis of 3 *Candida* species using an integrated and refined bioinformatics pipeline (Stroehlein et al., 2015). Firstly, the 3 proteome FASTA sequences were isolated from the UniProt database (<https://www.uniprot.org/protomes/?query=&sort=score>) and the data was uploaded into Kinanote program by choosing parameter non metazoan (Goldberg et al., 2013). The proteomes thus obtained were filtrated according to their sequences, which were further classified into group, family and or subfamily and a few remaining as uncharacterized proteins. The uncharacterized proteins, which were not recognized by Kinanote required manual curation. The uncharacterized proteins were individually subjected to BLAST (protein–protein) (<https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins&>) against the proteome of *Saccharomyces* for pairwise homologous sequence comparison. The fitted homologous sequences of model species were analyzed in UniProt (<https://www.uniprot.org/>) and subsequently added to a kinase group, family or subfamily.

2.2. Phylogenetic tree construction

We localized the kinase domains of all the curated kinase proteins of *C. albicans* by using the HMMER program (<http://hmm.janelia.org/>) and Interproscan (<https://www.ebi.ac.uk/interpro/search/sequence-search>) to construct individual kinase groups based on their kinase sequences. The kinase domain sequences of all predicted kinases were aligned using the program MAFFT server, employing the L-INS-i options

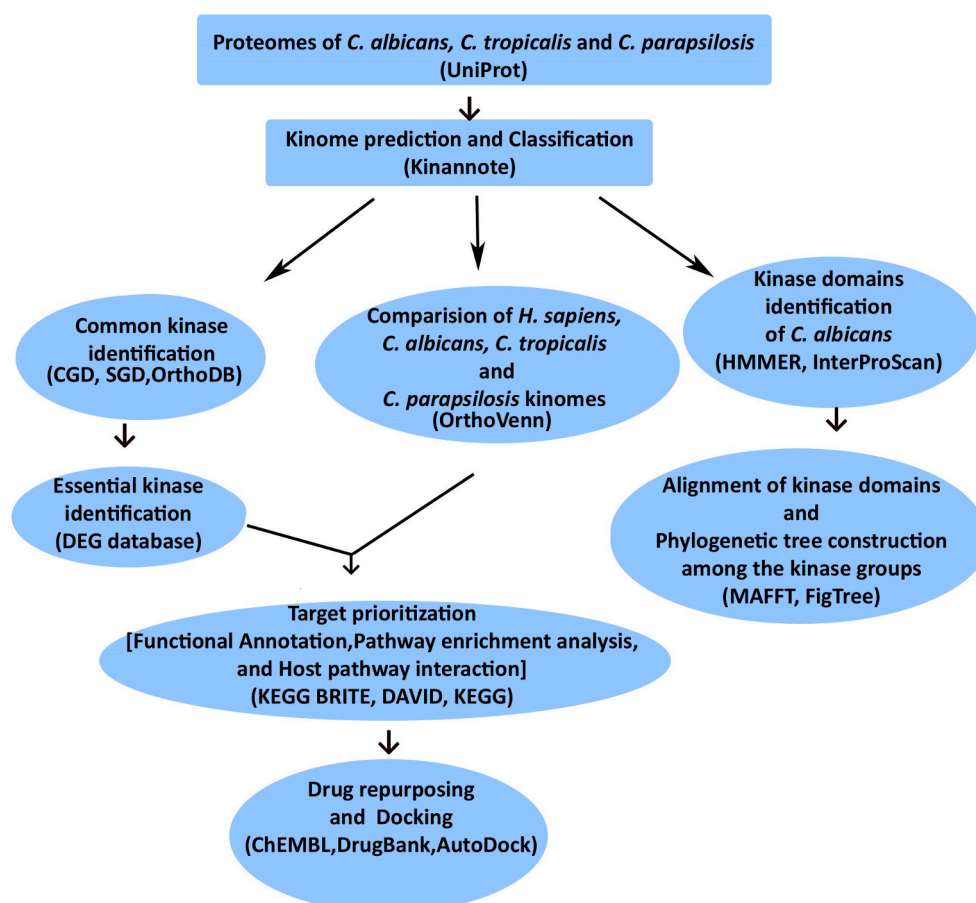


Fig. 1. Bioinformatics pipeline to curate and characterize kinases for target prioritization of candidiasis.

(Katoh et al., 2018). Phylogenetic tree of kinase groups were constructed with the aligned resultants by the program FigTree v1.4.3 (Rambaut, 2016).

2.3. Common kinase identification and prediction of essential proteins

The UniProt ID of all kinase proteins from the 3 species were searched individually in *Candida* genome database (<http://www.candidagenome.org/>), OrthoDB (<https://www.orthodb.org/>) and Saccharomyces Genome Database (SGD) (<https://www.yeastgenome.org/>) to retrieve the information about gene names of *C. albicans* and orthologous genes in other *Candida* spp. as well as in *S. cerevisiae*. From their gene names and or orthologous gene names we summarized the common *C. albicans* protein kinases. These common proteins were reposed on database of essential genes (DEG) and BLAST was performed by using BLASTp function against each essential protein sequences of *S. cerevisiae* (Luo et al., 2014; Zhang, 2004; Zhang and Lin, 2009). The parameter of shortlisting the essential genes were set at a minimum 100 bit score (>80% identity) with 0.0001 as expectation value (Barh et al., 2011).

2.4. Comparison of *C. albicans*, *C. parapolis*, *C. tropicalis* kinomes and homo sapiens proteins

The kinome sequences of all 3 *Candida* spp. and Human kinome (<http://kinase.com/kinbase/FastaFiles/>) were analyzed in OrthoVenn 2 to draw relativeness among them. Finally we identified the common kinase sequences present in all the *Candida* spp. but absent in human i.e., the human non-homologous proteins.

2.5. Functional annotation

Curated kinase sequences were functionally annotated by searching the databases KEGG BRITE (BlastKOALA) (database release May 15, 2019) using query amino acid sequences in FASTA format and fungi as taxonomy group (Kanehisa et al., 2016) and Gene Ontology database (<http://geneontology.org/>) was used for further curation.

2.6. Pathway enrichment analysis

To analyze the integrated interactions between kinase proteins and metabolic pathways, the common selected essential human non-homologous proteins were mapped on the metabolic pathways using DAVID server (Huang et al., 2009a, 2009b).

2.7. Host pathway interaction study

To identify the kinase proteins, which are not involved in the pathways of other organisms and uniquely present in *Candida*, we used KEGG Genome Database for the comparison of metabolic pathways (http://www.genome.jp/kaas-bin/kaas_main) (Hossain et al., 2013; Moriya et al., 2007). Finally from the distinct *Candida* metabolic pathways we identified human non-homologous essential unique kinase proteins.

2.8. Three dimensional structure development of prioritized kinase proteins

The human non-orthologous but essential protein kinases of *Candida* species were identified and prioritized. The prioritized protein sequences were used in Swiss Model server (<https://swissmodel.expasy.org/interactive>) to generate three dimensional structures. Among the generated models, the first one of each protein was used in Galaxy Refiner for structural refinement (Ko et al., 2012). Further, proteins with multiple chains were separated in Chimera for drug prediction analysis (Waterhouse et al., 2009). Active sites and druggable pockets of these kinase proteins were explored with bioinformatics server ProBiS

(<http://probis.cmm.ki.si/>) (Konc et al., 2015).

2.9. Compound selection for docking evaluation

These prioritized kinases were aligned to homologous kinase sequences in the ChEMBL and DrugBank databases (Gaulton et al., 2012; Wishart et al., 2018). In DrugBank, sequence similarity search was performed using PSI-BLAST, in which protein type was filtered with the target and the enzyme. The resultant highest bit score similar sequences of protein drugs were curated on the basis of the activity, excluding inducer and activator. Also in ChEMBL the sequence similarity search of the prioritized kinases was done. The drug activity of the top most bit score protein was filtered according to standard type - Inhibition, IC₅₀, EC₅₀ with literature validation and also following the Lipinski rule-of-five (Lipinski, 2004). Finally 20 drugs from ChEMBL were chosen on the basis of inhibition percentage and IC₅₀ concentration. Three dimensional structures of all the curated drugs from DrugBank and ChEMBL were generated by SMILE through online smile translator (<https://cactus.nci.nih.gov/translate/>). A literature study was carried out in PubMed and PubChem to detect compounds of prioritized target kinases, which have not yet been evaluated against *Candida* species. After that the selected compounds were docked with their respective target proteins to detect efficacy of the drug towards its target proteins. AutoDock tools and Autodock Vina were used for docking (Oleg and Arthur, 2010). The docking was performed in the identified druggable pocket of each individual kinase generated from ProBiS server (<http://probis.cmm.ki.si>). The specific grid center (X,Y,Z) and grid size (X,Y,Z) on the three dimensional structure of the prioritized kinases were detected from the druggable pocket for AutoDock Vina (Supplementary-1). Further, the docking results were analyzed at Pymol (Schrödinger, 2015)

3. Result

The kinome analysis is a crucial approach towards the understanding and identification of patho-virulent kinase proteins and subsequently their possible drug targets. In this work, we identified all the kinomes from the 3 different pathogenic *Candida* spp. and orthologous common kinases among them. We also focused on functions and pathways of these kinases to draw the attention on important drug targets.

3.1. Definition and prediction of *C. albicans*, *C. parapolis* and *C. tropicalis* kinomes

For this we collected proteomes of all the *Candida* spp. from UniProt and generated 103, 106 and 103 kinases of *C. albicans*, *C. parapsilosis* and *C. tropicalis*, respectively in Kinannotate (Supplementary-2). After manual curation of the uncharacterized proteins from Kinannotate, 3 proteins (CTRG_03116, CTRG_02191, CTRG_02511) were added into the kinome of *C. tropicalis* and 1 (CPAR2_206880) protein in *C. parapsilosis*, so the total kinome were 103, 107 and 106 in *C. albicans*, *C. parapsilosis* and *C. tropicalis*, respectively. These kinase proteins of the 3 species are distributed in 5 well defined groups and a group containing diverse protein family i.e Other group under ePK and another atypical group. (Table 1)

Total 103 protein kinases of *C. albicans* were assembled into 6 ePK groups (including others group) and an atypical kinase group. AGC group contained 12 protein kinases distributed into 6 families and 1 protein remained as unique AGC kinase. CAMK group had 16 kinases classified into 3 families and 4 proteins were uniquely present as CAMK kinase. CK1 group had 3 kinases present only in cell kinase 1 family. The 23 protein kinases of CMGC group were assigned to 8 families and 1 as unique CMGC kinase. Among the STE group, 12 kinases were classified into 3 families. Other group contained 26 protein kinases belonging to 18 different families. 4 proteins remain as unassigned serine/ threonine kinase (STK) family. Along with these ePKs, there were 7 atypical

Table 1
Classification of Kinases into groups and families.

Group	Family	<i>C. albicans</i>	<i>C. parapsilosis</i>	<i>C. tropicalis</i>
AGC	AKT	2	2	2
	NDR	2	2	2
	PDK1	2	2	2
	PKA	2	2	2
	PKC	1	1	1
	RSK	2	2	3
	Unique	1	1	
CAMK	AGC			
	Unique	4	3	3
	CAMK			
	CAMK1	3	4	4
	CAMKL	7	7	7
CK1	RAD53	2	2	2
	CK1	3	3	4
CMGC	CDK	6	7	7
	CK2	2	2	2
	CLK	1	1	1
	DYRK/	2	2	2
	DYRK2			
	GSK	3	2	2
	MAPK	5	5	5
	RCK	1	1	1
	SRPK	2	2	2
	unique	1		
	CMGC			
STE	STE11	4	4	4
	STE20/	5	5	5
	PAKA			
	STE7	3	3	3
UnassignedSerine/		4	4	4
threonine Kinase				
(STK)*				
Others	HAL	5	6	6
	family			
	AUR	1	1	1
	BUB	1	1	1
	BUD32	1	1	1
	CDC7	1	1	1
	NAK	3	2	3
	IKS	1	1	1
	IRE	1	1	1
	NEK	1	1	1
	PEK/	1	1	1
	GCN2			
	PLK	1	1	1
	RAN	1	1	1
	SCY1	2	2	2
	TTK	1	1	1
	ULK	1	1	1
	VPS15	1	1	1
	WEE	1	1	1
	CAMKK	2	2	2
Atypical	ABC1	3	4	4
	HisK	1	1	1
	PIKK	1	2	1
	PIPK			1
	PIK4		1	1
	RIO	2	2	2
	PI3/PI4		1	

*Few proteins are unassigned to any of the ePK groups which are place in table as unassigned STK

protein kinases classified into 4 families.

In *C. parapsilosis* a total of 107 protein kinase representing 6 ePK groups (including others group) and also atypical kinase group were identified. The largest group CMGC represented 22 protein kinases sparsely distributed in 8 different families. The second largest group is CAMK representing 16 kinases were included in 3 different families and 3 proteins uniquely remain as CAMK kinase. AGC group has 12 protein kinases distributed in 6 families and one unique AGC kinase. CK1 group had 3 kinases present in only CK1 family. There are 12 protein kinases in STE group, included in 3 families. Among the others groups, 26 protein

kinase were including in 18 families. 4 proteins remain as unassigned to serine/ threonine kinase (STK) family. Beside these ePKs, there were 11 atypical protein kinases classified into 6 families.

In *C. tropicalis* a total of 106 protein kinases were identified, of which there were representatives of 6 ePK kinase groups (including other group) and an atypical kinase group. The group CMGC has 22 protein kinases classified into 8 families. CAMK group representing 16 kinases were included in 3 families and 3 proteins were grouped as unique CAMK kinase. There were 12 protein kinases grouped in AGC, classified into 6 families. CK1 group had 4 kinases present in only CK1 family. Among the STE group, 12 protein kinases were distributed in 3 families. Other group has 26 protein kinases, which belong to 17 different families. 4 proteins remain as unassigned to serine/ threonine kinase (STK) family. Along with these ePKs, there were 10 atypical protein kinases classified into 6 protein kinase families.

3.2. Molecular phylogenetics of kinomes

The whole kinome of *C. albicans* comprising of 103 proteins were analyzed by HMMER, Interproscan and were finally localized in the kinase domain. The sequences of these kinase regions of all proteins were aligned at MAFFT. The results were analyzed in FigTree v1.4.3, which built the phylogenetic relationship among the different kinase proteins and their groups (Fig. 2.). In the phylogenetic tree, CMGC group shows maximum number of clade, which suggests that proteins of this group were sparsely distributed in maximum number of families, whereas CK1 group having lowest number of clades and proteins belong to only 1 family, which were also similarly obtained from Kinannote.

3.3. Common kinases and essential proteins in *Candida* spp.

From the said *Candida* genome database, OrthoDB and Saccharomyces Genome Database (SGD), we identified 79 kinase proteins of *C. albicans* which have orthologous kinases in other two species *C. parapsilosis* and *C. tropicalis* (Supplementary-3). Among these 79 common kinases, 54 are essential, short listed by stipulated parameter in database of essential genes (Supplementary-4).

3.4. Comparison of *C. albicans*, *C. parapsilosis*, *C. tropicalis* kinomes and *homo sapiens* proteins

The kinomes of all 3 *Candida* spp. and the human were compared in order to identify the candidate drug targets among the pathogens. The reason behind this approach is that the kinases of *Candida* spp. that cluster together and that do not cluster with the human kinases may be considered as plausible targets. By using the kinase protein sequences of the *C. albicans*, *C. parapsilosis*, *C. tropicalis* and *H. sapiens* kinomes, in OrthoVenn was infer the orthologous clusters of kinases (Fig. 3). There were 51 clusters of common orthologous sequences for all the 4 species and 42 proteins clusters only in the 3 *Candida* spp. These 42 proteins were considered as human non-homologues. These 42 human non-homologous protein kinases were compared with 54 essential proteins generated from database of essential gene and eventually 9 essential human non-homologous proteins were identified.

3.5. Functional annotations of essential and human non-homologous kinases

The functional importance of 9 essential human non-homologous protein kinases was analyzed in KEGG BRITE. 7 proteins were annotated, of which 3 proteins Cdc15, Pkh3, Hsl1 showed environmental information processing, Rim15 kinase was involved in cellular processing, Kin2 has role in metabolism and Ptk2, orf19.3854 (CAALFM_CR06040WA) proteins showed unclassified metabolisms (Fig. 4.). The gene ontology study shows several statistically significant results among the molecular functions and biological processes,

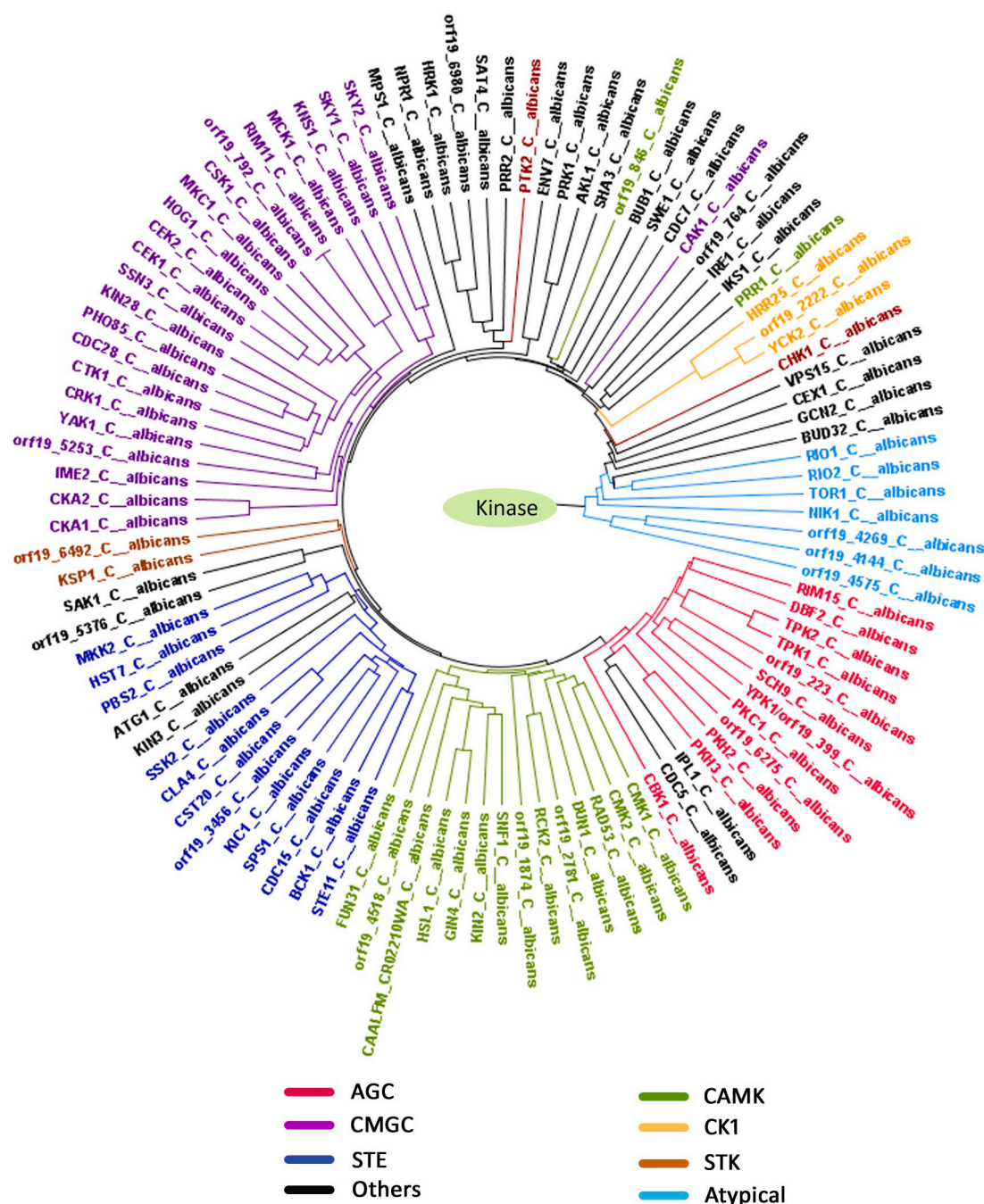


Fig. 2. Phylogenetic analysis of the protein kinase groups of *Candida albicans*. The kinase domains of *C. albicans* were used to construct the tree. Each of the groups is highlighted in different colours.

whereas, no significant cellular component is found (Table 2).

3.6. Pathway enrichment analysis of essential and human non-homologous kinases

The pathway involvement essentialities of the 9 prioritized kinases outlined their necessity in *Candida*. The results of DAVID for pathway enrichment analysis shows Hsl1 is involved in cell cycle (cal04111) regulation, Cdc15 also regulates cell cycle (cal04111) as well as meiosis (cal04113) process and Rim15 regulate the meiosis (cal04113) pathway in *C. albicans*. Whereas, Pkh3 is involved in 8 signaling pathways including, inflammation mediated by chemokine and cytokine signaling pathway (P00031), insulin/IGF pathway-protein kinase B signaling cascade (P00033), interleukin signaling pathway (P00036), PDGF

signaling pathway (P00047), PI3 kinase pathway (P00048), p53 pathway (P00059), Ras Pathway (P04393) and p53 pathway feedback loops 2 (P04398) (Table 3).

3.7. Host pathway interaction analysis of essential and human non-homologous kinases

The host pathway interaction analysis was done to avoid the cross species sensitivity. In this study, among the 9 non-homologous essential proteins, 4 proteins Hsl1, Rim15, Pkh3 (PDPK1), Cdc15 are involved in 29 metabolic pathways in multiple organisms, including 4C. *albicans* (yeast) pathways. Hsl1 is the only protein which is present in 2 *Candida* (yeast) pathways out of 29 metabolic pathways (Table 4). So, the remaining 5 essential non-homologous proteins, Npr1, Kin2, orf19.3854

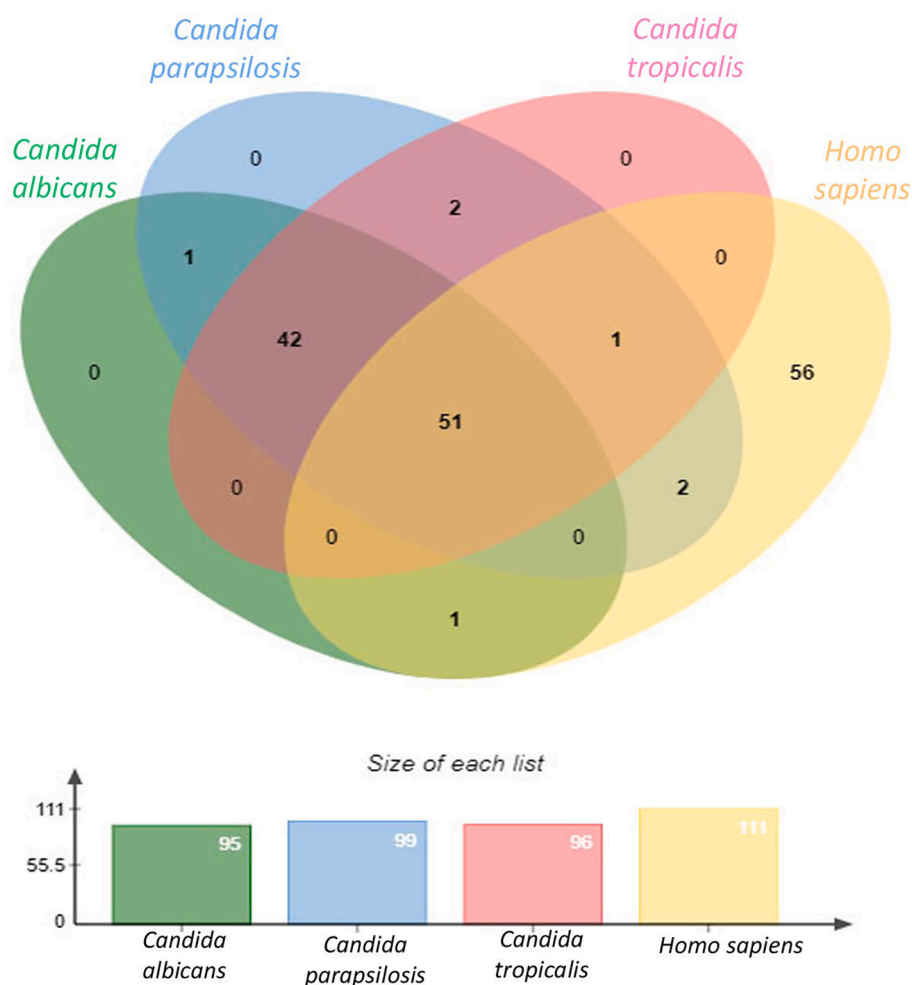


Fig. 3. Venn diagram depicting the clusters of orthologous kinases in *C. albicans*; *C. parapsilosis*; *C. tropicalis* and *H. sapiens*.

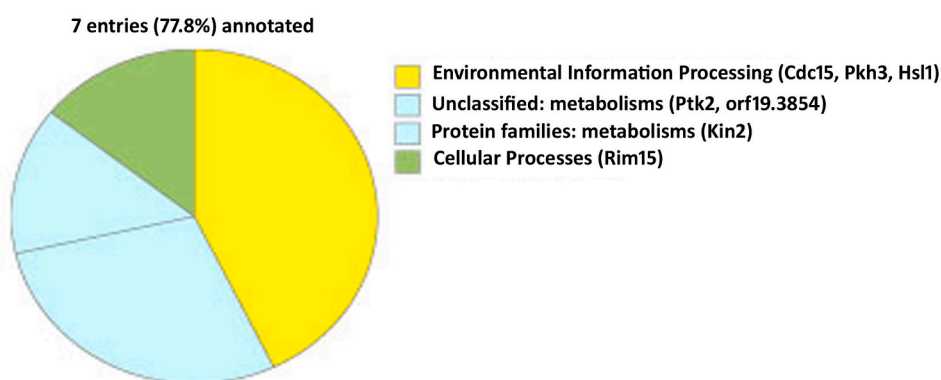


Fig. 4. Functional annotation result of prioritized essential human non-homologous kinases.

(CAALFM_CR06040WA), Ksp1, Ptk2 and uniquely involved in *Candida*, Hsl1 protein may be considered as the promising therapeutic targets in *Candida*.

3.8. Drug target prioritization and docking evaluation of selected compounds

The curated 6 human non-homologous essential kinase protein 3D structures were built and their structural corrections were done. Two proteins Hsl1 and orf19.3854 (CAALFM_CR06040WA), have 2 chains,

which were split for further drug target predictions. All the druggable pockets and active sites of the proteins were identified where ligand docking was performed. Individually all the proteins were docked with their respective predicted drug compounds (Supplementary-5). The results were screened according to their virtual docking binding free energy to establish the binding affinity between the target protein receptors and drug ligands. The compounds with more negative binding free energy value were suggested to be more effective as the target molecule (Hopkins and Paolini, 2007) and the binding interaction of the top three compounds against each of prioritized kinase were presented

Table 2

Functional annotation of the prioritized kinases.

Molecular function	Gene	P-value
protein serine/threonine kinase activity (GO:0004674)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.80E-16
protein kinase activity (GO:0004672)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	9.23E-16
protein tyrosine kinase activity (GO:0004713)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	3.82E-15
phosphotransferase activity, alcohol group as acceptor (GO:0016773)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.26E-14
kinase activity (GO:0016301)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	4.57E-14
transferase activity, transferring phosphorus-containing groups (GO:0016772)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.69E-12
catalytic activity, acting on a protein (GO:0140096)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	1.52E-10
ATP binding (GO:0005524)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	6.72E-10
adenyl ribonucleotide binding (GO:0032559)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	7.57E-10
adenyl nucleotide binding (GO:0030554)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	8.32E-10
drug binding (GO:0008144)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	3.04E-09
purine ribonucleoside triphosphate binding (GO:0035639)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	3.29E-09
purine ribonucleotide binding (GO:0032555)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	3.70E-09
purine nucleotide binding (GO:0017076)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	4.16E-09
ribonucleotide binding (GO:0032553)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	5.41E-09
carbohydrate derivative binding (GO:0097367)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	6.04E-09
transferase activity (GO:0016740)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	1.39E-08
nucleotide binding (GO:0000166)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	4.74E-08
nucleoside phosphate binding (GO:1901265)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	4.74E-08
anion binding (GO:0043168)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	8.04E-08
small molecule binding (GO:0036094)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	1.12E-07
ion binding (GO:0043167)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	2.79E-06
heterocyclic compound binding (GO:1901363)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	4.11E-05
organic cyclic compound binding (GO:0097159)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, KIN2	4.35E-05
Biological process complete		
protein phosphorylation (GO:0006468)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	6.47E-16
intracellular signal transduction (GO:0035556)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.15E-14
phosphorylation (GO:0016310)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	5.00E-14
signal transduction (GO:0007165)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	4.11E-13
signaling (GO:0023052)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	4.90E-13

Table 2 (continued)

Molecular function	Gene	P-value
peptidyl-serine phosphorylation (GO:0018105)	RIM15, PKH3, KSP1, NPR1, HSL1, CDC15, KIN2	6.43E-12
peptidyl-serine modification (GO:0018209)	RIM15, PKH3, KSP1, NPR1, HSL1, CDC15, KIN2	8.64E-12
cell communication (GO:0007154)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	1.83E-10
phosphate-containing compound metabolic process (GO:0006796)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.47E-10
phosphorus metabolic process (GO:0006793)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	3.30E-10
protein modification process (GO:0036211)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	5.14E-10
cellular protein modification process (GO:0006464)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	5.14E-10
macromolecule modification (GO:0043412)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	1.64E-09
peptidyl-amino acid modification (GO:0018193)	RIM15, PKH3, KSP1, NPR1, HSL1, CDC15, KIN2	4.09E-08
cellular protein metabolic process (GO:0044267)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	5.84E-08
protein metabolic process (GO:0019538)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	1.06E-07
cellular response to stimulus (GO:0051716)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	3.39E-07
response to stimulus (GO:0050896)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	9.18E-07
regulation of cellular process (GO:0050794)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	1.04E-06
regulation of biological process (GO:0050789)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.19E-06
organonitrogen compound metabolic process (GO:1901564)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.70E-06
cellular macromolecule metabolic process (GO:0044260)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.83E-06
protein autophosphorylation (GO:0046777)	RIM15, HSL1, KIN2	4.55E-06
biological regulation (GO:0065007)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	9.04E-06
macromolecule metabolic process (GO:0043170)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.25E-05
regulation of cell growth (GO:0001558)	KSP1, HSL1, KIN2	4.07E-05
mitotic cell cycle phase transition (GO:0044772)	HSL1, PTK2, CDC15	1.04E-04
nitrogen compound metabolic process (GO:0006807)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	1.07E-04
cell cycle phase transition (GO:0044770)	HSL1, PTK2, CDC16	1.09E-04
regulation of cellular component organization (GO:0051128)	RIM15, PKH3, HSL1, CDC15, KIN2	1.12E-04
cell cycle (GO:0007049)	RIM15, PKH3, HSL1, CDC15, KIN2	2.10E-04
primary metabolic process (GO:0044238)	RIM15, PKH3, KSP1, NPR1, HSL1, PTK2, CDC15, CAALFM_CR06040WA, KIN2	2.29E-04

Table 3Pathway involvement of the prioritized kinases in *Candida albicans*.

Gene Name	KEGG_PATHWAY
HSL1 (CaO19.11784; CaO19.4308) (Maglott et al., 2007)	cal04111:Cell cycle,
CDC15 (CaO19.3545) (Maglott et al., 2007)	cal04111:Cell cycle,cal04113:Meiosis,
RIM15(CaO19.7044) (Maglott et al., 2007)	cal04113:Meiosis,
PANTHER_PATHWAY	
PKH3 (CaO19.1196) (Maglott et al., 2007)	P00031:Inflammation mediated by chemokine and cytokine signaling pathway,P00033: Insulin/IGF pathway-protein kinase B signaling cascade,P00036:Interleukin signaling pathway, P00047:PDGF signaling pathway,P00048:PI3 kinase pathway,P00059:p53 pathway,P04393: Ras Pathway,P04398:p53 pathway feedback loops 2,

Table 4

Host pathway interaction of the prioritized kinases.

Sl. No	Pathway Name (Reference pathway(KO))	Protein involved (KO)
1	MAPK signaling pathway – yeast (ko04011)	Hsl1, PDPK1 (ortholog Pkh3)
2	Hippo signaling pathway - multiple species (ko04392)	Cdc15 (KO6683)
3	FoxO signaling pathway (ko04068)	PDPK1(ortholog Pkh3)
4	Sphingolipid signaling pathway (ko04071)	PDPK1(ortholog Pkh3)
5	PI3K-Akt signaling pathway (ko04151)	PDPK1(ortholog Pkh3)
6	AMPK signaling pathway (ko04152)	PDPK1(ortholog Pkh3)
7	mTOR signaling pathway (ko04150)	PDPK1(ortholog Pkh3)
8	Autophagy- animal (ko04140)	PDPK1(ortholog Pkh3)
9	Autophagy – yeast (ko04138)	Rim15
10	Cell cycle – yeast (ko04111)	Hsl1, Cdc15
11	Meiosis – yeast (ko04113)	Cdc15, Rim15
12	Apoptosis (ko04210)	PDPK1(ortholog Pkh3)
13	Focal adhesion (ko04510)	PDPK1(ortholog Pkh3)
14	T cell receptor signaling pathway (ko04660)	PDPK1(ortholog Pkh3)
15	Fc epsilon RI signaling pathway (ko04664)	PDPK1(ortholog Pkh3)
16	Insulin signaling pathway (ko04910)	PDPK1(ortholog Pkh3)
17	PPAR signaling pathway (ko03320)	PDPK1(ortholog Pkh3)
18	Thyroid hormone signaling pathway (ko04919)	PDPK1(ortholog Pkh3)
19	Aldosterone-regulated sodium reabsorption (ko04960)	PDPK1(ortholog Pkh3)
20	Neurotrophin signaling pathway (ko04722)	PDPK1(ortholog Pkh3)
21	Longevity regulating pathway - multiple species (ko04213)	Rim15 (K12767)
22	Proteoglycans in cancer (ko05205)	PDPK1(ortholog Pkh3)
23	Choline metabolism in cancer (ko05231)	PDPK1(ortholog Pkh3)
24	Prostate cancer (ko05215)	PDPK1(ortholog Pkh3)
25	Endometrial cancer (ko05213)	PDPK1(ortholog Pkh3)
26	Non-small cell lung cancer (ko05223)	PDPK1(ortholog Pkh3)
27	Insulin resistance (ko04931)	PDPK1(ortholog Pkh3)
28	Toxoplasmosis (ko05145)	PDPK1(ortholog Pkh3)
29	Platinum drug resistance (ko01524)	PDPK1(ortholog Pkh3)

(Fig. 5) (Table 5).

4. Discussion

The bioinformatics workflow through a definite pipeline enabled the first complete classification of the kinomes of 3 *Candida* species important to control human disease. The curation procedure revealed 103, 107 and 106 protein kinases in *C. albicans*, *C. parapsilosis* and *C. tropicalis* respectively, these numbers represent 1.70%, 1.85% and 1.70% of their respective genomes. The large number of kinase proteins in the genus suggests that phosphorylation is a very important process in its pathobiology. The differences in kinase numbers occur may be due to the intra-species variability, besides these 79 common kinase proteins were also identified. Out of these 79 common kinases, 54 proteins were

involved in essential functions in the pathogen. Our intent was to select the human non-homologous proteins for the target prioritization analysis, for which we studied the sequence homology and revealed that half of the common kinase proteins were not clustered with human. Since decades, the expedition into biological targets of pathogens with homologous in mammals has been kept away and it aims at eliminating the possible problems of selectivity and its adverse effects during the process of drug development. From these approaches, we identified 9 kinase proteins which are essential to *C. albicans* and do not have human homologues. The functional annotation of these essential human non-homologous proteins, revealed that 3 proteins were involved in environmental information processing, 1 was in metabolism, 2 proteins were involved in unclassified metabolisms and 1 belongs to cellular processes. The gene ontology study of these 9 proteins showed various biological activities mostly molecular functions and biological processes, and no significant cellular component was found. Within the molecular functions, 8 proteins showed drug binding (GO:0008144) activity with significant p-value 3.04E-09, which revealed that these kinases would be a good therapeutic targets. Additionally, all the protein kinases apart from protein phosphorylation (GO:0006468) are involved in intracellular signal transduction (GO:0035556), which is one of the most important biological processes. The pathway enrichment analysis showed that 4 proteins were involved in cell cycle (cal04111), meiosis (cal04113), p53 pathway (P00059) and p53 pathway feedback loops 2 (P04398), which were significant and established cell survival pathways. From the curated 9 essential human non-homologous kinases, 3 were associated with various host pathways and remaining 6 kinase proteins Hsl1, Npr1, Kin2, orf19.3854 (CAALFM_CR06040WA), Ksp1, Ptk2 eventually become promising therapeutic targets. The Hsl1 protein is a key enzyme playing central role in cell cycle control and virulent yeast to hyphae morphogenesis. Hsl1 suppress the cell elongation process of cell cycle progression as a controlling factor of Swe1-Cdc28 pathway. The disruption of the Hsl1 also shows reduced pathogenicity in kidney colonization in mouse systemic infection (Umeyama et al., 2005). This study postulated Hsl1 as one of the important kinase, involved in several molecular and biological functions. Additionally Hsl1 is involved in MAPK signaling pathway and cell cycle regulation, along with drug binding affinity. Drug skimming revealed ChEMBL3745885; ChEMBL560075 and ChEMBL2333365 are the most promising drugs against this kinase. Npr1 is a core stress response gene important for Mep2 mediated ammonium transport signaling which induces yeast to filamentous morphogenesis in *C. albicans* on low nitrogen condition (Biswas and Morschhäuser, 2005; Enjalbert et al., 2006; Neuhaus et al., 2011). Npr1 belongs to HAL family kinase involved in different molecular functions notably showing ATP binding, drug binding and small protein binding affinity. Npr1 also significantly regulates various biological processes such as intracellular signal transduction; protein phosphorylation; cell communication; cellular response to stimulus and nitrogen compound metabolic process. n-[5-[4-(4-methylpiperazin-1-yl)phenyl]-1h-pyrrolo[2,3-b]pyridin-3-yl]nicotinamide; 4-(6-[(4-methylcyclohexyl)amino]methyl)-1,4-dihydroindeno[1,2-c]pyrazol-3-yl]benzoic and Enzastaurin are the most probable promising drugs against the Npr1, which we curated. Ptk2 is required for efficient polyamine import, and raised levels significantly contribute to the biofilm phenotype (García-Sánchez et al., 2004; Kaouass et al., 1997; Nishikawa et al., 2016; Nozaki et al., 1996). Ptk2 deletion increases yeast cell wall thickness; β -glucan exposure and resistance to calcoflour white and hygromycin (Garcia et al., 2018). In this study, Ptk2 probably plays a role in various important biological and molecular processes in the organisms. Mitotic cell cycle phase transition is one of the important biological phenomena probably regulated by Ptk2. The drug mining outcome of the Ptk2 protein shows ChEMBL1163566; ChEMBL3745885; ChEMBL1164181 compounds has the lowest binding free energy. Kin2 is involved in biofilm development (Konstantinidou and Morrissey, 2015) and is also involved in cell wall stress sensitivity phenotype (Blankenship et al., 2010). In this study Kin2 probably

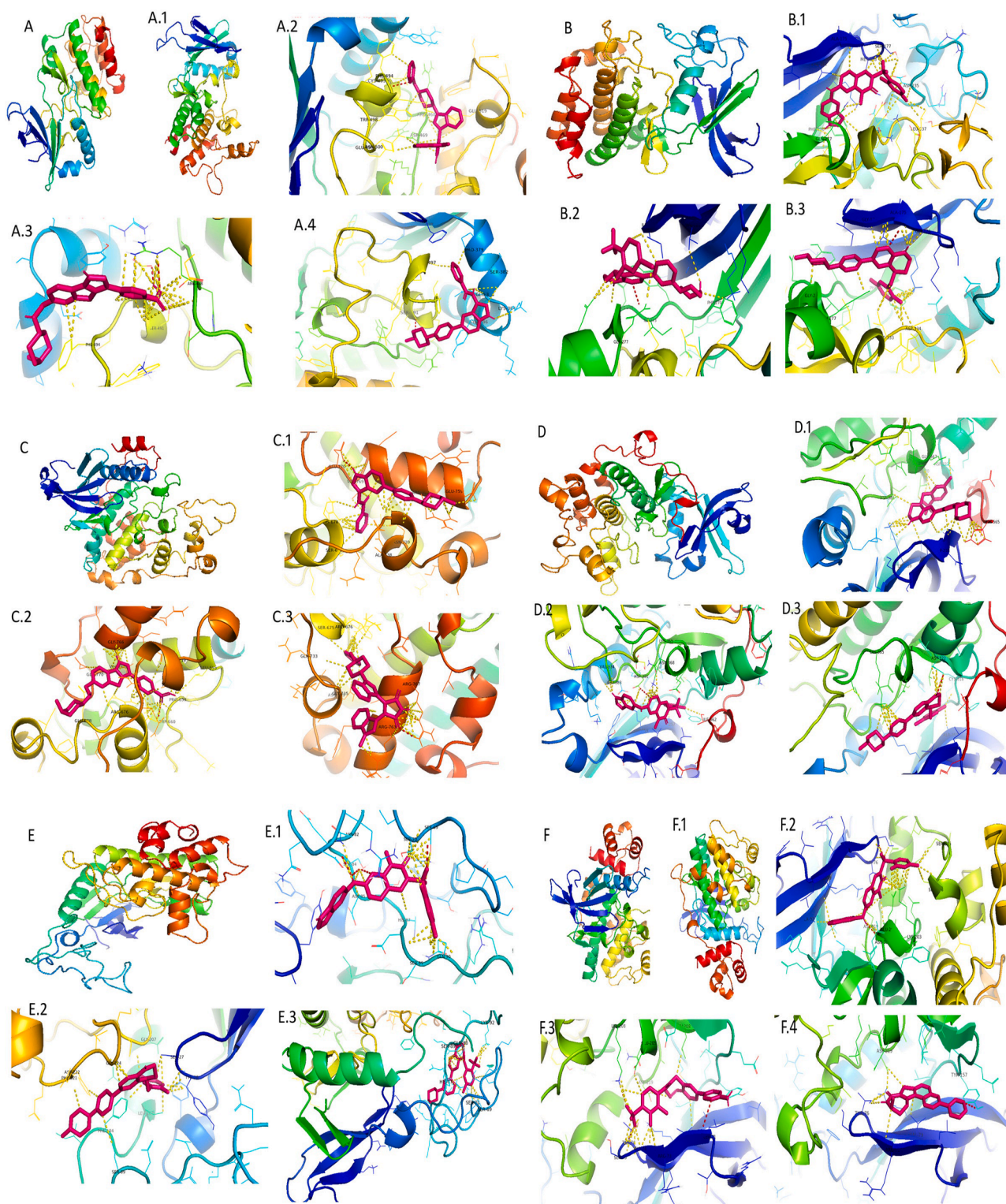


Fig. 5. Prioritized proteins and their lowest free binding affinity interactions with drugs. A - orf19.3854 Chain-A; A.1- orf19.3854 Chain-B; A.2- orf19.3854 Chain-A and Enzastaurin interaction; A.3- orf19.3854 Chain-A and 4-(6-((4-methylcyclohexyl)amino)methyl)-1,4-dihydroindeno[1,2-c]pyrazol-3-yl)benzoic acid interaction ; A.4 - orf19.3854 Chain-A and n-{5[4-(4-methylpiperazin-1-yl)phenyl]-1h-pyrrolo[2,3-b]pyridin-3-yl}nicotinamide interaction. B- Kin2 protein; B.1- Kin2 protein and CHEMBL3745885 interaction; B.2- Kin2 protein and CHEMBL1163566 interaction; B.3 - Kin2 protein and CHEMBL1165499 interaction. C - Npr1 protein; C.1 - Npr1 protein and n-{5[4-(4-methylpiperazin-1-yl)phenyl]-1h-pyrrolo[2,3-b]pyridin-3-yl}nicotinamide interaction; C.2 - Npr1 protein and 4-(6-((4-methylcyclohexyl)amino)methyl)-1,4-dihydroindeno[1,2-c]pyrazol-3-yl)benzoic acid interaction; C.3 - Npr1 and Enzastaurin interaction. D - Ptk2 protein; D.1 - Ptk2 and CHEMBL1163566 interaction; D.2 - Ptk2 and CHEMBL3745885 interaction; D.3 - Ptk2 and CHEMBL1164181 interaction. E - Ksp1 protein; E.1 - Ksp1 and CHEMBL1165499 interaction; E.2 - Ksp1 and CHEMBL1163566 interaction; E.3 - Ksp1 and CHEMBL1163566 interaction. F - Hsl1 Chain-A; F.1 - Hsl1 Chain-B; F.2 - Hsl1 Chain-B and CHEMBL3745885 interaction; F.3 - Hsl1 Chain-B and CHEMBL233365 interaction; F.4 - Hsl1Chain-A and CHEMBL560075 interaction. All interactions are showed within 3.5Å and polar interactions are marked in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

List of chemical compounds as drug candidates against candidiasis.

Kinase proteins	orf19.3854 (CAALFM_CR06040WA)	Npr1 (kcal/mol)	Ptk2(kcal/ mol)	Ksp1 (kcal/ mol)	Kin2 (kcal/ mol)	Hsl1	
Drugs	Chain A (kcal/mol)	Chain B (kcal/mol)				Chain A (kcal/mol)	Chain B (kcal/mol)
CHEMBL3745885							
n-{5[4-(4-methylpiperazin-1-yl)phenyl]-1h-pyrrolo [2,3-b]pyridin-3-yl}nicotinamide	−9.3		−9.7	−9.3	−9.6	−9.5	−8.7
4-(6-[[4-(4-methylcyclohexyl)amino]methyl]-1,4- dihydroindeno[1,2-c]pyrazol-3-yl)benzoic acid	−9.4		−9.5				
Enzastaurin		−10.1	−9.2				
CHEMBL1163566				−10.6	−8.7	−9.2	
CHEMBL1164181				−9.1			
CHEMBL2333365							−8.6
CHEMBL1165499				−8.9	−9.1		
CHEMBL560075						−8.6	

participates in several molecular and biological processes. The cell growth regulation; cell communication; auto-phosphorylation and drug binding affinity make the kinase as potent drug target molecule. We found after docking methods, CHEMBL3745885; CHEMBL1163566 and CHEMBL1165499 drugs have the potentiality to act on Kin2 protein. The Ksp1 protein contributes in nutrient stress-responsive signaling as a TORC1 and PKA pathway downstream effector that regulates hyphal growth (Cutler et al., 2001; Mutlu et al., 2019). Ksp1 as a part of the PKA signaling network, negatively regulates autophagy (Umekawa and Klionsky, 2012). Ksp1 mutation confers hypersensitivity to amphotericin B (Xu et al., 2007). Ksp1 is probably involved in 24 molecular functions as well as cellular processes. Our drug mining investigation against Ksp1, showed that CHEMBL3745885; CHEMBL1163566 and CHEMBL1165499 will be the probable therapeutic molecule. orf19.3854 (CAALFM_CR06040WA) belongs to the HAL protein kinase family; induces spider biofilm formation (Nobile et al., 2012). During amphotericin B exposure its expression is increased (Liu et al., 2005). This protein is less studied in *C. albicans*, and is probably involved in 7 molecular and numerous biological processes. 4-(6-[[4-(4-methylcyclohexyl)amino]methyl]-1,4-dihydroindeno[1,2-c]pyrazol-3-yl)benzoic acid; n-{5[4-(4-methylpiperazin-1-yl)phenyl]-1h-pyrrolo[2,3-b]pyridin-3-yl}nicotinamide and Enzastaurin are the potent probable drugs against this kinase.

An extensive study of literature related to kinase profiling and cytotoxicity of the drug compounds has been performed. CHEMBL3745885 is an arylsulfonyle pyridopyrimidinone, well tolerated in mice, IC₅₀ was 0.075 μM in both K562, DU145 cell lines and kinase profiling against the 355 functional protein kinases, at high concentration shows inhibitory activity (Reddy et al., 2016). The IC₅₀ value of CHEMBL1163566 compound was studied in Karpas-299 cell which was reported as 150 nM (Milkiewicz et al., 2010). The cytotoxicity of the compound CHEMBL2333365 was studied 50 mg/kg in mice model which did not show any significant adverse effect, also kinase profiling among the 141 kinases showed IC₅₀ values > 3000 nM (Takeuchi et al., 2013). The kinase profiling of the compound CHEMBL560075 among the 50 kinases showed only 4 kinases have > 70% inhibition at 1 μM (Anderson et al., 2009). The kinase selectivity of the compound CHEMBL1163566, CHEMBL1164181 and CHEMBL1165499 at concentration of 1 μM showed only 6.7%, 2%, and 1.2% of the evaluated kinases were inhibited > 90% respectively (Milkiewicz et al., 2010). The compound n-{5[4-(4-methylpiperazin-1-yl)phenyl]-1h-pyrrolo[2,3-b]pyridin-3-yl}nicotinamide belongs to the class of organic compounds known as phenylpiperazines. The organic phenylpyrazoles compound 4-(6-[[4-(4-methylcyclohexyl)amino]methyl]-1,4-dihydroindeno[1,2-c]pyrazol-3-yl)benzoic acid has two isoforms and inhibit Chk1 human kinase at IC₅₀ of 20 nM and 22 nM (Tong et al., 2007). Enzastaurin is a piperidiny derivative, exhibits IC₅₀ at 0.7 μM in rat brain Pkc and human Pkc varies between 8 and 0.03 μM depending on various isozymes. Human Pkc β isozyme shows lowest IC₅₀ value 0.03 μM. The

Kinase selectivity assay showed the compound is > 300-fold more selective for Pkc β relative to calcium calmodulin and > 3000-fold more selective for src-tyrosine kinase and Casein kinase (Faul et al., 2003). Therefore, the extensive study on kinase profiling and toxicity of the therapeutic compounds reveal that most of them are probable good clinical candidates for therapeutic trial against candidiasis.

5. Conclusion

In conclusion, thorough bioinformatics workflow led to the classification of *C. albicans*, *C. parapsilosis* and *C. tropicalis* kinomes into majority of the kinases at family levels. All these kinase proteins were annotated according to their functions and compared with human homologues. Finally 6 proteins were predicted to be good drug targets that will be useful for treating candidiasis. The drug repurposing pipeline revealed the kinase inhibitor molecules CHEMBL1163566 and CHEMBL3745885, probably as promising drugs against 3 and 4 protein kinases respectively. Enzastaurin; n-{5[4-(4-methylpiperazin-1-yl)phenyl]-1h-pyrrolo[2,3-b]pyridin-3-yl}nicotinamide and 4-(6-[[4-(4-methylcyclohexyl)amino]methyl]-1,4-dihydroindeno[1,2-c]pyrazol-3-yl)benzoic showed promising results against the 2 proteins Npr1 and orf19.3854 (CAALFM_CR06040WA). The compound CHEMBL1165499 is also a promising candidate for 2 proteins Ksp1, Kin2. Compounds CHEMBL1164181; CHEMBL2333365 and CHEMBL560075 have the potential to be therapeutic targets against single kinase protein. The cytotoxicity and the kinase profiling of the compounds were previously studied in various cell lines, animal models and in different assay conditions. The compounds CHEMBL3745885, CHEMBL1163566, CHEMBL2333365 and Enzastaurin have been studied for both cytotoxic and kinase inhibitory profiling. The rest of the compounds require investigation in terms of cytotoxicity and or kinase inhibitory profiling. Further future investigations of prioritized 6 kinases are also needed in order to study their roles in pathogenesis, establish the efficacy and species specificity of kinase inhibitors, experimentally in KO strains and also in vivo candidiasis model.

CRedit authorship contribution statement

Sanjib Das: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Rajabrata Bhuyan:** Methodology, Software. **Achintya Mohan Goswami:** Conceptualization, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - review & editing. **Tanima Saha:** Conceptualization, Funding acquisition, Investigation, Methodology, Resources, Project administration, Software, Supervision, Validation, Visualization, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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